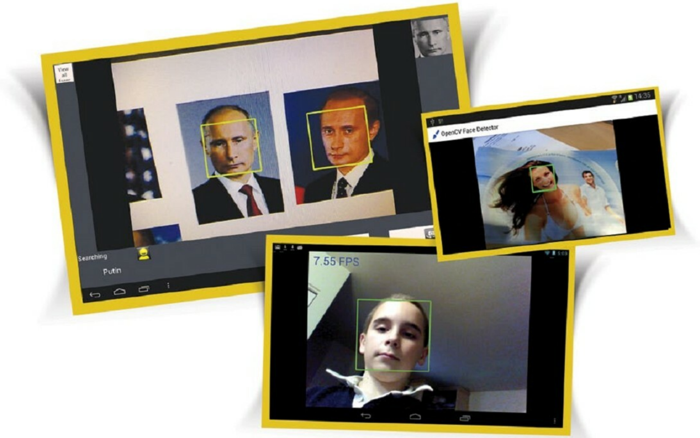
OpenCV4Android Make Your Smartphones See

OpenCV is a computer library that is primarily aimed at real-time computer vision. Now that mobile devices come with powerful cameras and high processing power, mobile platforms are ideal for executing computer vision programs.



OpenCV is the most popular open source computer vision library. Computer vision deals with tasks like image reception, processing and understanding of visual content, and has applications in a large number of fields ranging from robotics to solid state physics. If you are an absolute beginner in this field, the December 2013 issue of *OSFY* has an introductory article on the topic. There are umpteen resources available on the Web to learn about computer vision and the OpenCV library. The documentation provided in the official OpenCV site itself is comprehensive and includes the reference manual, user guide and tutorial sections. The popularity and utility of OpenCV can be judged by the number of downloads, which are more than nine million to date.

OpenCV platforms

The OpenCV versions are available for all major operating systems like Windows, Linux, Mac and FreeBSD. Primarily, OpenCV is written in the C++ language and hence can be easily ported across various environments. OpenCV support is also extended to CUDA (Compute Unified Device Architecture), which is a parallel computing platform and programming model by NVIDIA. This is directed towards tapping the potential of graphics processing units (GPU). Studies have proved that by better utilising the resources of the GPU, primitive image processing tasks can be speeded up by 30 times compared to using the CPU alone.

Mobile operating systems like Android, Maemo and iOS also support OpenCV. This article is about the Android port of OpenCV, which is termed OpenCV4Android.

OpenCV4Android

Nowadays, smartphones have become extremely powerful with respect to their hardware profiles. The presence of a camera and multi-core microprocessor facilities in smartphones has made them capable of performing computer vision tasks. Needless to say, their portability enhances their applicability in this domain.